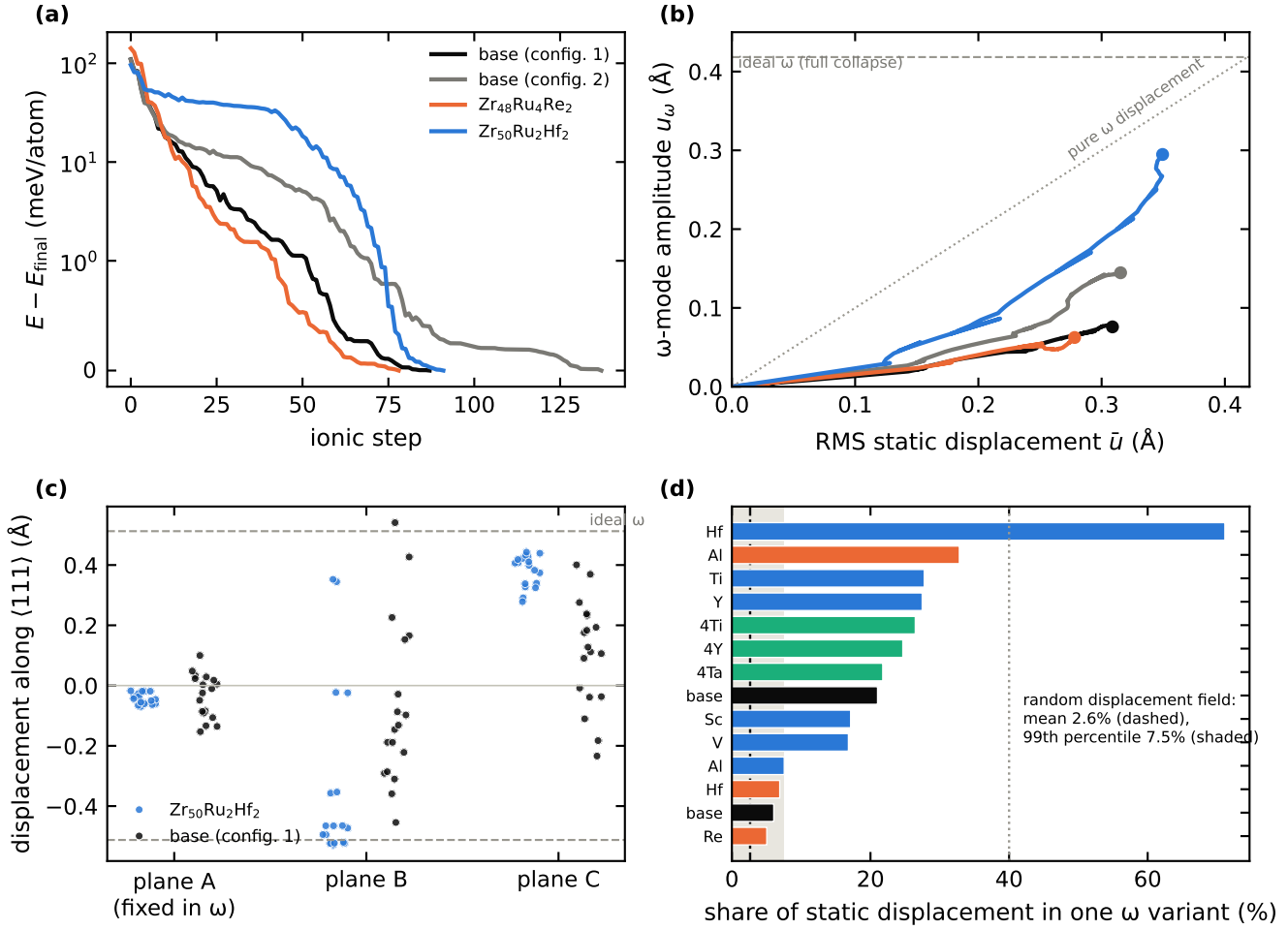
(a)
base (config. 1)
base (config. 2)
Zr48Ru4Re2
Zr50Ru2Hf2
E − Efinal (meV/atom)
ionic step
(b)
ideal ω (full collapse)
pure ω displacement
ω-mode amplitude uω (Å)
RMS static displacement ū (Å)
(c)
ideal ω
displacement along ⟨111⟩ (Å)
Zr50Ru2Hf2
base (config. 1)
plane A
(fixed in ω)
plane B
plane C
(d)
Hf
Al
Ti
Y
4Ti
4Y
4Ta
base
Sc
V
Al
Hf
base
Re
random displacement field:
mean 2.6% (dashed),
99th percentile 7.5% (shaded)
share of static displacement in one ω variant (%)